

2021 Ph H2 Q11

Section: Particles and Waves

Topic: The Bohr Model of the Atom

Analogy: books on shelves in a bookcase used to represent the Bohr model of the atom. Comment on the strengths and limitations of this analogy.

Worked solution

- In the Bohr model, electrons are restricted to discrete energy levels. Similarly, books are restricted to certain shelves (not between shelves).
- Movement of an electron between levels involves the absorption or emission of a photon. Analogously, a book can be moved between shelves, but it cannot rest at an in-between height.
- However, the analogy is limited: books can stay on any shelf indefinitely, whereas electrons have specific rules for allowed transitions and quantum states.
- Also, the book analogy does not capture the idea of quantised angular momentum, nor the probabilistic nature of quantum mechanics.

Final points

The analogy is useful because it shows electrons only occupy discrete levels and must jump

between them. It is limited because it ignores angular momentum quantisation and quantum probabilities.

Revision tips

- Bohr model: electrons in fixed energy levels (quantised).
- Photon absorbed/emitted when electron changes level.
- Analogies: shelves/books help visualise discreteness but have limits.
- Real atoms follow quantum mechanics, not simple orbits.
- Use analogies carefully—highlight both strengths and weaknesses.